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Transition Metal-Oxide High Entropy Microstructure for Alkaline Hydrogen Evolution Reaction

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Abstract

Hydrogen (H_2) is widely recognized as a sustainable, non-polluting renewable fuel. Despite significant progress in hydrogen evolution reaction (HER), fundamental understanding of the reaction mechanisms and structure–activity relationships under alkaline conditions remains incomplete. Furthermore, the development of catalysts that exhibit high activity, durability, and stability under realistic operating conditions is essential for the large-scale deployment of hydrogen production technologies. In this context, studying the HER under alkaline conditions is not only of academic interest but also of practical importance for future hydrogen-based energy systems, including alkaline electrolyzers. Alkaline water electrolysis is a promising technology for sustainable hydrogen production, where water gets split into hydrogen and oxygen using electricity in an alkaline medium. Alkaline water electrolyzers have advantages over acidic systems, as they allow the use of non-precious metal catalysts, making them more cost-effective. By leveraging renewable electricity, alkaline water splitting can yield green hydrogen, a clean fuel that emits only water when used, making it a viable alternative for future energy needs. In this work, a high entropy metal oxide electrocatalyst has been synthesized via simple ball milling technique. The catalyst powder was thoroughly characterized, and its activity was evaluated for HER in alkaline medium. Furthermore, to explore the practical use, accelerated durability test was performed for 500 cycles, where it showed high stability.

Introduction

The transition toward a sustainable energy future demands innovative solutions to meet the growing global energy needs while reducing environmental impact. Hydrogen is an ideal, sustainable and renewable clean fuel due to its high mass-based energy density and environmental friendliness. Electrocatalytic water splitting, as a key technology for the future powered by H_2 , has attracted the interest of both industries and academics due to its effectiveness, economy, low-carbon, and cleanliness.^{1,2} The global H_2 demand increased by 3% in 2022 and reached 95 million tons. Almost all H_2 used in industry is produced from fossil fuel sources, such as natural gas and coal. The hydrogen generation and utilization in 2022 caused about 900 Mt of CO_2 emissions. The amount of green H_2 produced by water splitting (0.1% of today's global

H₂ generation) and blue H₂ produced from fossil fuel-based sources with carbon capture, utilization, and storage is only about 0.7%. Alkaline water electrolysis (AWE) is one of the most practical and advanced technologies with a long history of utilization in the chlor-alkali industry for large-scale H₂ production. The AWE has begun to gain more attention due to the increasing demand and interest in green H₂. This exponential growth and interest are forcing academia and industry to revisit and improve some of its aspects, especially the development of a low-cost and stable electrocatalyst. Unlike acidic water electrolysis, AWE enables the use of earth-abundant, non-noble metal electrocatalysts, offering a more economically viable and scalable alternative.

Advanced functional materials such as nanostructured catalysts, hybrid composites, and low-dimensional materials offer routes to overcome the limitations of conventional hydrogen production and energy conversion technologies. In water electrolysis, for instance, highly active and earth-abundant electrocatalysts can replace scarce noble metals, drastically lowering costs while enhancing hydrogen evolution reaction (HER). HER is a critical half-cell reaction in water electrolysis, playing a pivotal role in sustainable hydrogen production. The activity of the electrocatalyst depends on several physicochemical properties of the material, including the composition, conductivity, electronic and crystal structures, morphology and textural parameters, as well as the preparation method, grain boundaries, surface structure, and the presence of defects. The performance of an electrocatalyst can be boosted either by increasing the number of active sites on a given electrode, through enhanced loading or altering structural properties to expose more catalytic active sites per gram or manipulating the intrinsic activity of each active site by keeping the mass constant. These advanced functional materials not only push the boundaries of performance but also address critical sustainability and scalability challenges. Continued research and development in this field will be essential to unlocking the full potential of hydrogen as a future fuel and enabling a clean er, greener energy landscape for generations to come. Recent research has focused on the rational design of advanced electrocatalysts for alkaline HER by leveraging a variety of material classes, including transition metal-based compounds (e.g., oxides, phosphides, sulfides, nitrides), metallic alloys, heterostructures, and even single-atom catalysts.³ Strategies to overcome the kinetic limitations involve enhancing intrinsic activity through electronic structure tuning, creating bifunctional active sites to separately catalyze water dissociation and hydrogen evolution, and engineering the catalyst surface to increase the electrochemically active area and promote efficient mass and electron transport.

High-entropy materials (HEMs), as a new type of materials, containing five or more elemental compositions and single-phase properties, have attracted extensive attention in the field of photocatalysis, electrocatalysis, and supercapacitors due to their unique structure, high chemical activity and component tunability. Unlike conventional metal compounds, the high number of elemental species in HEMs makes them provide at least five binding energies, and the random distribution of the different elements can cause the atoms of the same element to exhibit varied binding energies due to their different local environments.^{4,5} Therefore, this broadens the distribution of binding energies to a continuous state that can provide catalytic sites with different binding energies for different intermediates, thus facilitating activity. High Entropy Alloy Oxides (HEAOs) which are the oxide analogue of HEAs, also show interesting electrochemical properties with high stability under alkaline water electrolysis condition. The unique structural and compositional complexity of HEAOs leads to a rich landscape of active sites with tunable electronic structures, potentially enabling multiple catalytic functionalities in a single material system. Here we have synthesized transition metal oxide high entropy oxide via mechanical alloying such as ball milling⁶ and the activity for alkaline hydrogen evolution reaction has been tested in 1M aqueous KOH electrolyte solution.

Experimental

Materials

Vanadium oxide (VO₂), chromium oxide (Cr₂O₃), manganese oxide (MnO₂), iron oxide (Fe₃O₄), cobalt oxide (Co₃O₄) are for synthesis. Potassium hydroxide (KOH), carbon black powder, 2-propanol, 5 wt% aqueous Nafion are for electrocatalysis reaction. All the chemicals were obtained commercially and used without any purification.

Synthesis of high entropy oxide solid solution

Transition metal oxides including vanadium oxide (VO₂), chromium oxide (Cr₂O₃), manganese oxide, iron oxide (Fe₃O₄), cobalt oxide (Co₃O₄) are mixed in an appropriate molar ratio and ball milled for at least 48 hours in a 100 ml stainless steel jar with a single stainless steel grinding ball. After the completion of the ball milling, the powder was collected and labelled as HEMO. This catalyst powder was used for characterization and electrochemical testing.

Characterization

X-ray Diffraction (XRD) patterns of the samples were recorded on a SmartLab Rigaku diffractometer using CuK α radiation ($\lambda=1.5406$ Å). X-ray photoelectron spectroscopy (XPS) analysis was done using AlK α radiation (1486.71 eV) as the X-ray source and the data are acquired using Thermo Scientific Nexsa G2 X-Ray Photoelectron Spectrometer. Pass energies of 200 eV and 20 eV were used for survey scans and high resolution detailed scans, respectively. Scanning Electron Microscopy (SEM) of the catalyst was done using JEOL IT800 SHL microscope. Electrochemical testing was conducted using Autolab PGSTAT302N potentiostat/galvanostat workstation.

Electrocatalytic activity measurement

Water electrolysis activity for hydrogen evolution reaction (HER) was done in three-electrode set up with 1M KOH electrolyte medium. Glassy carbon electrode (area: 0.071 cm²) with drop-casted catalyst ink, Pt wire and Hg/HgO were used as working, counter and reference electrode, respectively. For that purpose, catalyst ink was prepared by dispersing 2mg catalyst powder in 960 μ L of 2- propanol and sonicated for 30 minutes followed by addition of 40 μ L 5wt% aqueous Nafion solution. This final mixture was sonicated for another 30 minutes to get homogeneous dispersion. Finally, 5 μ L catalyst ink was drop-casted on glassy carbon electrode and left for at least 6 hours to dry. The final catalyst loading was 0.14 mg cm⁻². Activity assessments were done by recording linear sweep voltammogram (LSV) with a scan rate of 5 mV s⁻¹. Electrochemical Impedance spectroscopy (EIS) analysis was performed within a frequency range of 0.1-10⁵ Hz at a particular potential. For accelerated durability test (ADT) was conducted by performing at least 500 cyclic voltammetry (CV) cycles with a scan rate of 50 mV s⁻¹. All the potentials ($E_{\text{Hg/HgO}}$) recorded with respect to Hg/HgO reference electrode were converted to reversible hydrogen electrode (E_{RHE}) by using the Nernst equation

$$E_{\text{RHE}} = E_{\text{Hg/HgO}} + E_{\text{Hg/HgO}}^{\circ} + 0.059\text{pH}$$

Here pH is 14 for 1M KOH, $E_{\text{Hg/HgO}}^{\circ}$ the standard potential of Hg/HgO (1M KOH) electrode and the value is 0.098V.

Results and Discussion

Mechanical alloying is an easy, conventional way to prepare alloy compound. It offers a robust, scalable, and solvent-free solid-state approach for synthesizing complex oxides through intensive energy input and repeated mechanical deformation. Mechanical ball milling is a top-down, solid-state powder processing

technique widely employed for the synthesis of high-entropy alloys (HEAs). This facilitates alloying at the atomic level, even for systems that are thermodynamically immiscible in conventional casting routes. Moreover, scaling up of this process is also possible for large scale synthesis and application. The energy input during milling facilitates cationic homogenization and may also promote solid-state reactions forming a single-phase oxide with high configurational entropy, even at room temperature. The phase of the ball milled powder was determined by XRD analysis. Figure 1a shows the XRD pattern of HEMO, suggesting presence of mixed oxide phases mostly hexagonal structure.

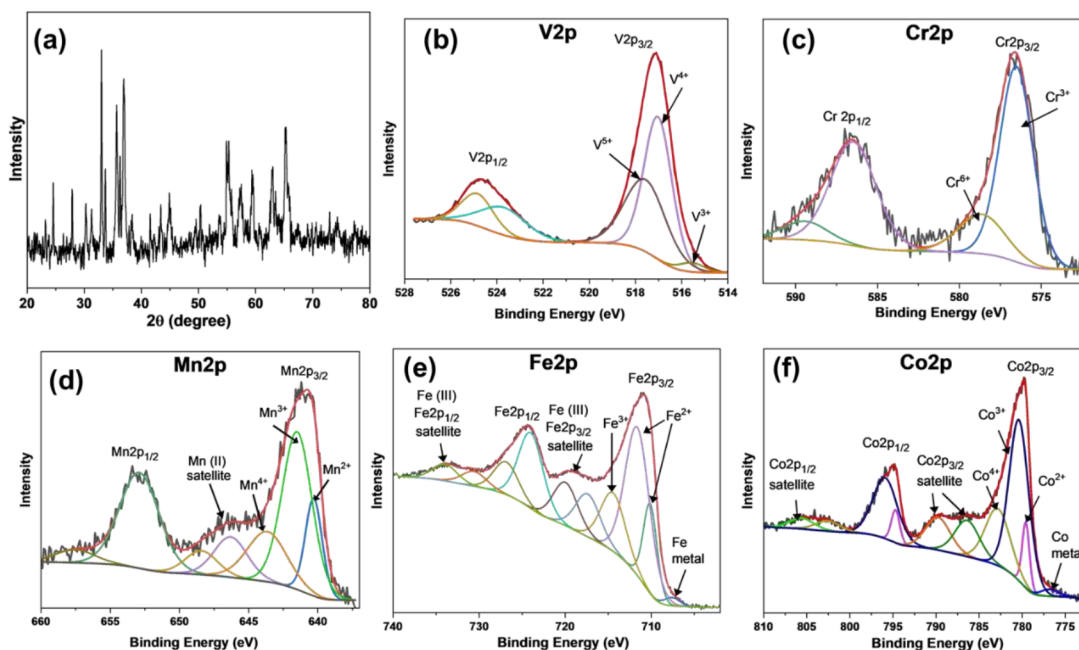


Figure 1—Structural analysis of HEMO: (a) XRD of HEMO, (b-f) XPS analysis of the elements present in HEMO.

A detailed XPS analysis of the catalyst powder was conducted to understand the chemical composition of the catalyst powder. Deconvoluted high resolution XPS data are shown in Figure 1b-f. V2p spectrum (Figure 1b) showed peaks related to V2p_{3/2} and V2p_{1/2} at ~517.1 and 524.7 eV, respectively. Deconvolution of the peaks revealed the presence of V³⁺, V⁴⁺ and V⁵⁺ oxidation states.⁷ Among these chemical states, V⁴⁺ was found to be present as major phase. The core level Cr2p spectrum shown in Figure 1c can be referred to the oxide peaks.⁸ The deconvolution of Cr2p spectrum gave rise to two individual peaks corresponding to Cr³⁺ and Cr⁴⁺ oxide.⁸ The major component was found to be Cr³⁺. Core level Mn2p (Figure 1d) shows two major peaks at a binding energy value of 640.8 and 652.9 eV correspond to Mn2p_{3/2} and Mn2p_{1/2}, respectively. Another shoulder peak at 646.8 eV can be assigned to the satellite peak generating from Mn(II) oxidation state. Deconvolution of Mn2p spectrum revealed the presence of Mn²⁺, Mn³⁺ and Mn⁴⁺ oxidation states. Figure 1e shows the core level Fe2p spectrum where presence of satellite shoulder peaks can be seen along with the Fe2p_{3/2} and Fe2p_{1/2} structural peaks. After deconvoluting the peaks, major contribution was found from Fe²⁺. For Co2p core level spectrum (Figure 1f), Co³⁺ was found to be the major contributing oxidation state of cobalt.⁹ Moreover, a minor metallic phase was also observed for Fe and Co. Therefore, the analysis confirmed that the catalyst synthesized is in oxide phase.

Morphology of the HEMO catalyst powder was investigated using SEM. The SEM image presented in Figure 2a shows the agglomerated nanoparticle morphology and the particle size was found to ~50-60 nm. To confirm the composition and distribution of the elements, EDS mapping was undertaken. Figure 2b shows the overlayed image and the individual elemental V, Cr, Mn, Co, Fe distribution can be found in Figure 2c-h, where homogenous distribution can be observed. Distribution of oxygen (Figure 2c) along

with metals confirmed oxide phase. The corresponding EDAX spectrum is shown in Figure 2i. Therefore, we have successfully prepared an alloy oxide microstructure.

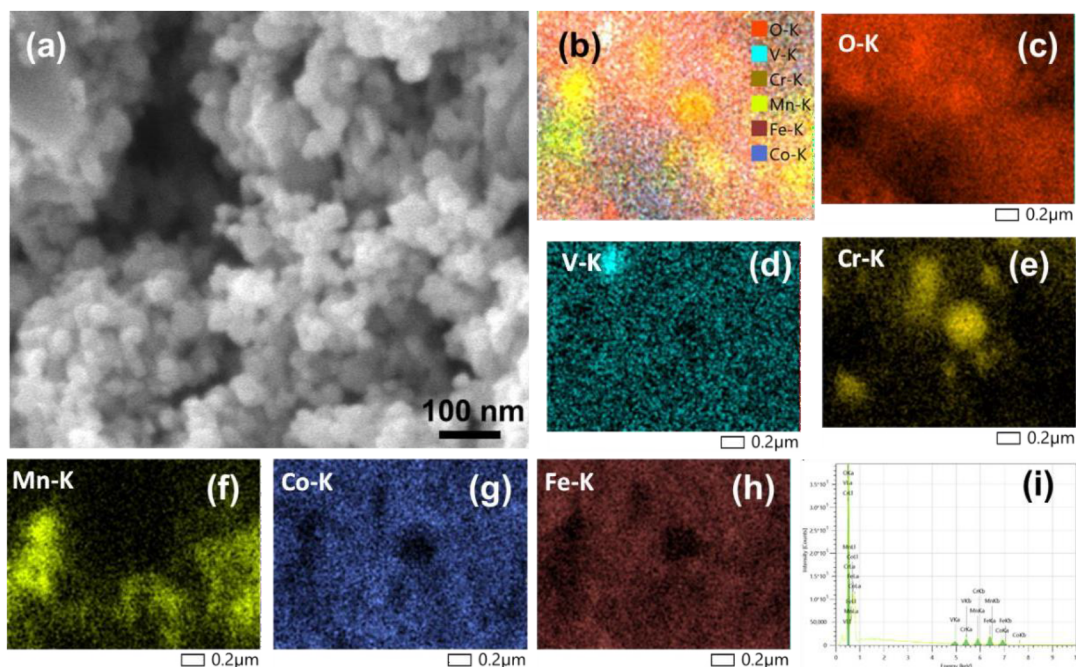


Figure 2—SEM analysis of HEMO: (a) SEM image, (b-h) EDS elemental mapping of the elements present in HEMO, (i) EDS spectrum from the mapping.

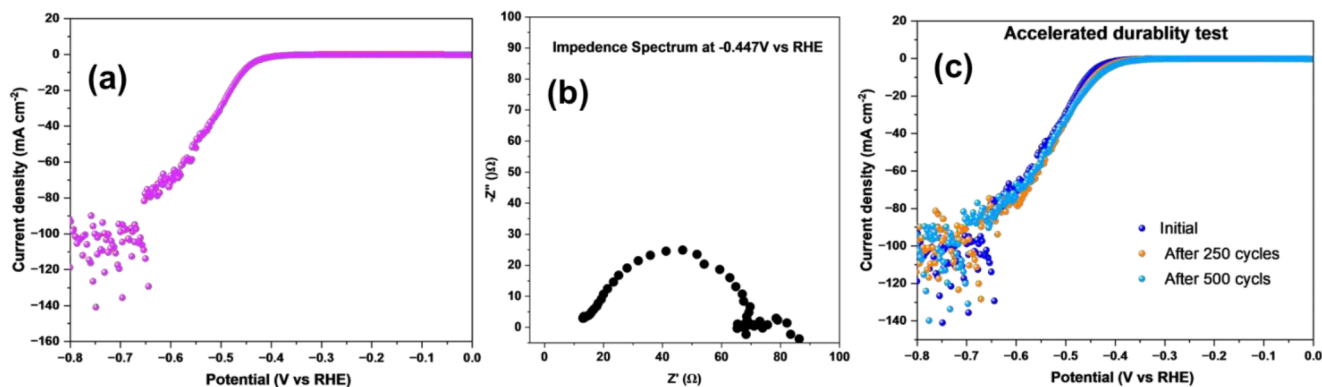
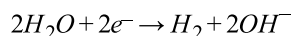


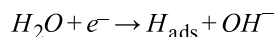
Figure 3—Electrocatalytic activity tests: (a) LSV for HER, (b) EIS at the overpotential ($\eta@10\text{mA cm}^{-2}$), (c) Accelerated durability test of HEMO.

Electrocatalytic activity of the HEMO for HER was investigated with 1M KOH. The overall hydrogen evolution reaction can be presented as



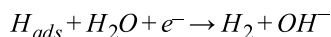
The HER in alkaline electrolyte (typically KOH) proceeds via the following steps on the catalyst surface:

1. Volmer step (electrochemical water dissociation):

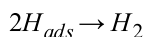


Here, a water molecule is reduced to adsorbed hydrogen (H_{ads}) and hydroxide ion (OH^-). This step is generally considered rate-determining step due to the energy barrier associated with breaking the H-OH bond.

- Heyrovsky step (electrochemical desorption):



- Tafel step (chemical recombination):



Linear Sweep Voltammetry (LSV) was performed to test the activity of HEMO for HER. Figure 3a shows the LSV recorded under a potential window of 0.0 to -0.8 V vs RHE. The overpotential ($\eta@10\text{mA cm}^{-2}$) is calculated by the following equation $E_{HER}^0 - E_{HER}^{Hg/HgO}$ and found to be -0.447 V vs RHE. Here E_{HER}^0 and $E_{HER}^0 - E_{HER}^{Hg/HgO}$ are the standard HER potential which is 0 V vs RHE, and potential required to achieve 10mA cm^{-2} current density value for the catalyst, respectively.

Electrochemical Impedance spectroscopy (EIS) analysis reveals the kinetics behind the electrochemical reaction by indicating different resistance involve on the process. To evaluate the charge transfer resistance, EIS was recorded at -0.447 V vs RHE and is presented in Figure 3b. The value was found to be $\sim 70 \Omega$, suggesting favorable kinetics. The activity of our HEMO catalyst was found to be as good as reported high entropy oxide with noble metal and noble metal based catalyst system.¹⁰⁻¹² Therefore, considering the cost effectiveness of the catalyst, our catalysts can be an alternative of high price noble metal based catalysts.

Accelerated durability test is one of the important studies to access the long-term stability of the catalyst. For that purpose, LSV data of HEMO catalyst were collected after 250 and 500 CV cycles conducted within a potential window of -0.3 to -0.45 V vs RHE. The corresponding data are shown in Figure 3c. No change in overall current density and $\eta@10\text{mA cm}^{-2}$ was observed as can be seen from Figure 3c. This result indicates structural stability of our catalyst under HER experimental conditions, therefore showing the potential for long term use. From the mechanism discussed earlier, it can be proposed that the oxide surface of our catalyst helps in adsorption of the active species, therefore facilitates the hydrogen evolution reaction.

Conclusion

In conclusion, a transitional metal based high entropy oxide microstructure has been synthesized through easily scalable, cost- effective mechanical ball milling approach. A detailed characterization using XRD, XPS and SEM confirmed homogenous distribution of the component elements and their presence in oxide form. As a result of synergistic effects of the nano-microstructure and combined effect resulted from the five elements, our catalyst showed HER activity in alkaline water electrolysis. It showed a low overpotential value of -0.447 V vs RHE at 10mA cm^{-2} current density values. While testing the stability, HEMO catalyst showed retention of its original activity even after 500 CV cycles. Such a stability test result suggests the potential of the newly developed catalyst for long-term real application. Therefore, this work holds a unique interest in the research area of energy conversion, especially hydrogen production.

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